

classical critical-path cost

condition that eliminates it

order log
on every message

network

Order: one global delivery order
→ replay without a log

coordinated snapshot
(markers + channel state)

Order + Barrier: uncoordinated
cut at timestamp T , channels empty

output held for extra
durability round trips

Barrier + Durability: durable inside
the barrier → no added round trip

local non-determinism
(time · randomness · threads)

host

Determinism: user-space shim
→ byte-identical replay